Lecture 6 2025-09-16

1.3 Algorithms and Convergence

Algorithm: A procedure to solve a problem (approximately) and convert an input into an output in a finite number of steps.

Pseudocode: Concise description of an algorithm, specifies form of input to be supplied and the desired form of the output.

Example: Evaluate $p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$

Input: $n, a_n, \ldots a_0, x$

Output: $p(x)$

Algorithm 1 Pseudocode:

Input: $n, a_0, \ldots, a_n, x$

Output: $p(x)$

Step 1: Set $p = a_0$, $x_j = 1$

Step 2: for $j = 1, \ldots, n$ do

$x_j = x_j * x$

$p = p + a_j * x_j$

assignment

return p

$x_j \leftarrow x_j * x$

$p \leftarrow p + a_j * x_j$

Alternatively, one can use Horner's rule (nested iteration)

$$p(x) = (\dots\dots(((a_n * x + a_{n-1})x + a_{n-2}) * x + a_{n-3})x + \dots\dots + a_1) * x + a_0$$

Pseudo code:

Input: $n, a_0, \dots, a_n, x$

Output: $p(x)$

Step 1: $p = a_n$

Step 2: for $j = n-1, n-2, \dots, 0$ do

$p = p * x + a_j$

return p

Requires half as many multiplications

Stability: Small changes in input data lead to small changes in output

Otherwise: unstable

3rd option: Algorithms that are stable for certain choices of input data are called conditionally stable

Def 1.17 Error propagation

E_0 = initial error

E_n = error after n operations

- If $E_n \leq C n E_0$ where C is a constant independent of n, then the growth of the error is linear.
- If $E_n \leq C^n E_0$ for some constant $C > 1$, then the growth is called

exponential.

Remark: Linear growth is usually unavoidable and when C and E_0 are small, it may be acceptable.
Exponential growth should be avoided because C^n becomes large very quickly.

Rates of convergence

Def 1.18 Let $\{\beta_n\}_{n=1}^{\infty}$ be a sequence known to converge to 0 and $\{\alpha_n\}_{n=1}^{\infty}$ a sequence converging to a value α.
If there exists a constant $K>0$ with

$$(\Delta) \qquad |\alpha_n - \alpha| \leq K|\beta_n| \quad \text{for large } n$$

then we say that $\{\alpha_n\}_{n=1}^{\infty}$ converges to α with *rate* or *order of convergence* $O(\beta_n)$

- One can write this statement also as $\alpha_n = \alpha + O(\beta_n)$
  "Big O Notation"

- Usually $\beta_n = \frac{1}{n^p}$ for some $p>0$. Then we are interested in the largest value p for which (Δ) holds. "*pth order convergence*"

- Even better if $\beta_n = \gamma^n \quad 0<\gamma<1$
  "*exponential convergence*"

Def 1.19 (for functions)
Suppose that $\lim_{h\to 0} G(h)=0$ and $\lim_{h\to 0} F(h) = L$
If a positive constant K exists with

$$|F(h)-L| \leq K|G(h)| \quad \text{for } h \text{ sufficiently small}$$

then we write $F(h) = L + O(G(h))$

Again usually we expect something like $G(h) = h^p$ for some $p > 0$

Example 2 from book

$$\alpha_n = \frac{n+1}{n^2} \quad , \quad \hat{\alpha}_n = \frac{n+3}{n^3}$$

$$\lim_{n\to\infty} \alpha_n = \lim_{n\to\infty} \hat{\alpha}_n = 0 \quad \text{but } \{\hat{\alpha}_n\} \text{ converges faster}$$

$$|\alpha_n - 0| = \frac{n+1}{n^2} \leq \frac{2n}{n^2} \leq \frac{2}{n} \Rightarrow \beta_n = \frac{1}{n} \quad K = 2$$

$(p = 1)$

$$|\hat{\alpha}_n - 0| = \frac{n+3}{n^3} \overset{n\geq 3}{\leq} \frac{2n}{n^3} = \frac{2}{n^2} \Rightarrow \beta_n = \frac{1}{n^2}, \; K = 2$$

$(p = 2)$

Example 3: Show that $\cos(h) + \frac{1}{2}h^2 = 1 + O(h^4)$ for small $h > 0$

Solution: Recall: $\cos(h) = 1 - \frac{1}{2}h^2 + \frac{1}{24}h^4 \cos(\xi(h))$

for $\xi(h) \in [0, h]$ (Taylor's theorem)

$$\left|\cos(h) + \tfrac{1}{2}h^2 - 1\right| = \frac{1}{24}h^4 \underbrace{|\cos(\xi(h))|}_{\leq 1} \leq \frac{h^4}{24}$$

$$K = \frac{1}{24}$$

$$G(h) = h^4$$

Chapter 2: Solving equations in one variable

Goal: Given $f(x)=0$ find x.

(find roots/zeros of f)

Root finding methods

2.1. Bisection Method

- Assume f is continuous on $[a,b]$ and $f(a)$ and $f(b)$ have opposite signs. By IVT there exists $p \in (a,b)$ s.t. $f(p)=0$

